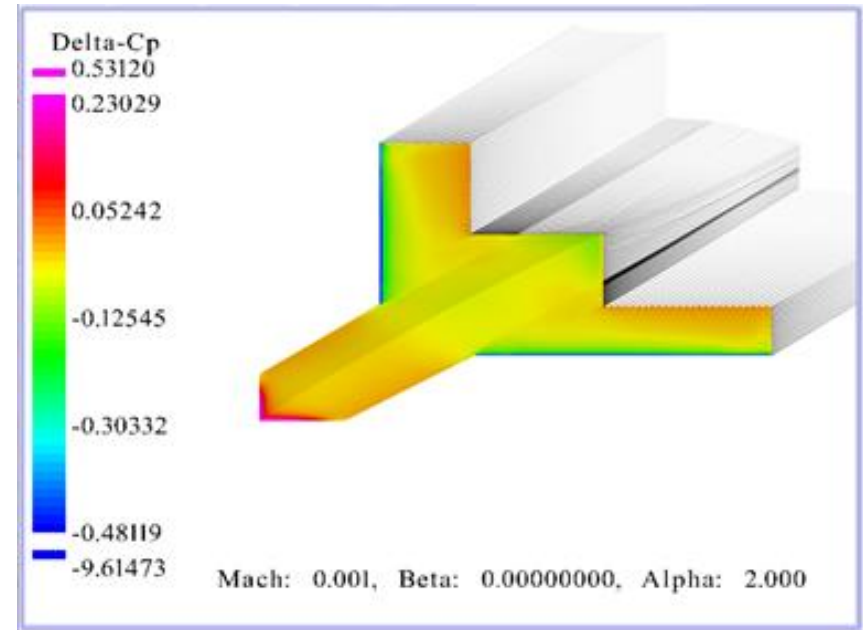
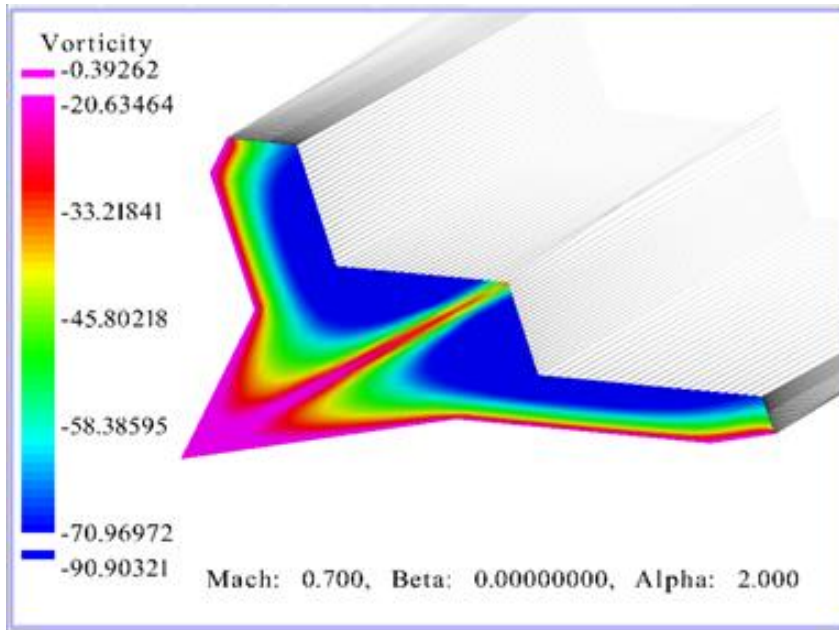


VSPAERO GUI & Tutorial - Basics



Presented by:

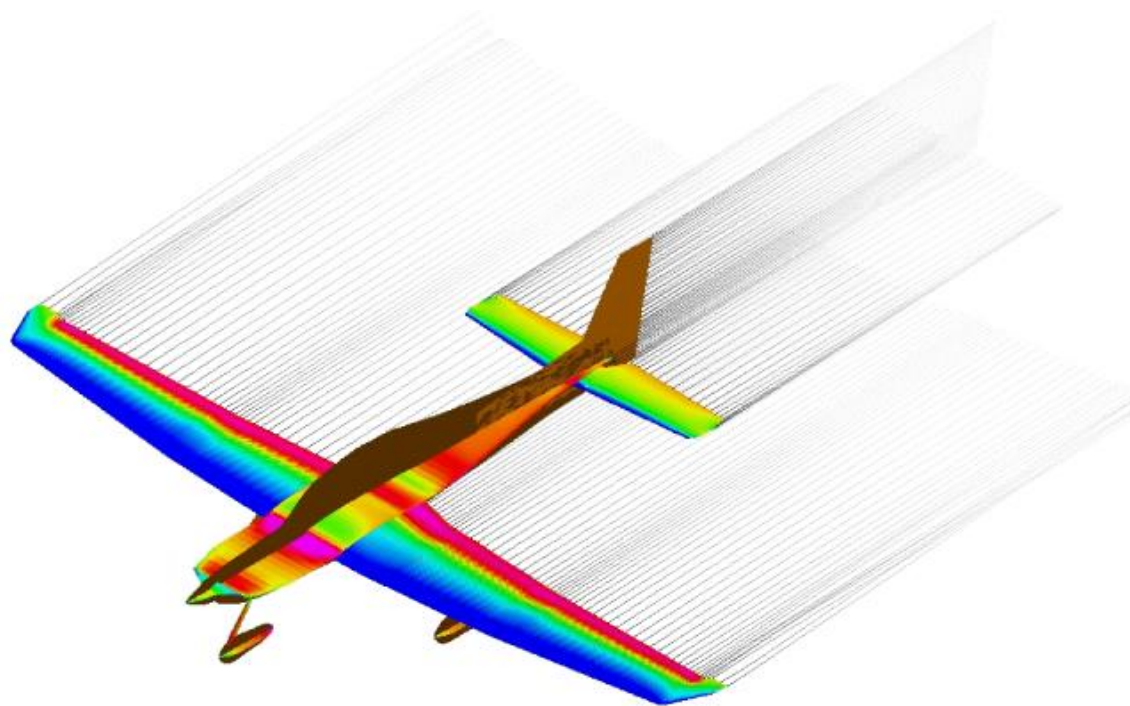
Justin Gravett

justin.gravett@esaero.com

Empirical Systems Aerospace, Inc. (ESAero)

Agenda

- Introduction
- VSPAERO GUI Walkthrough
- VSPAERO Results Manager
- VSP Viewer
- Exporting Results
- Basic Workflow
- Tessellation
- Modeling Tips for VSPAERO
- Demo



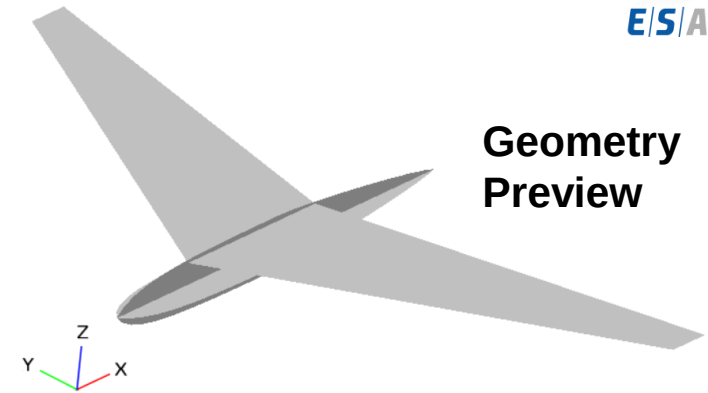
Introduction

Overview

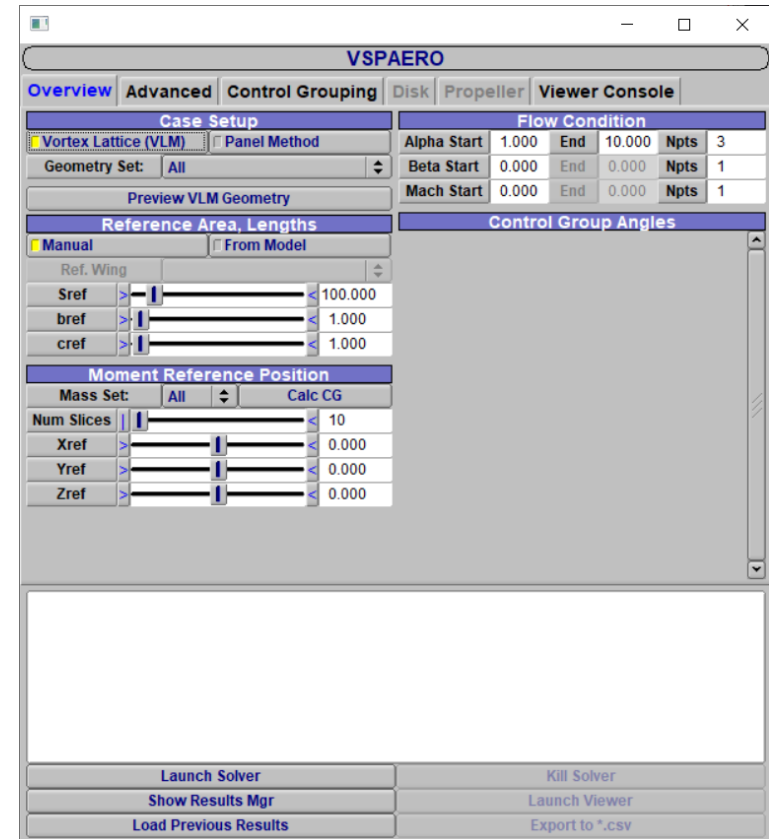
- VSPAERO and OpenVSP are separate programs
 - Interact through files and command line arguments
 - Slicer and Viewer are additional programs OpenVSP executes

Role of OpenVSP

- Geometric modeling tool
- Generate VSPAERO geometry input files (DegenGeom & Cart3D)
- User-friendly GUI to setup and run VSPAERO
- Collect and organize VSPAERO results
- Support API automation



Geometry Preview



GUI – Overview Tab (1/2)

Case Setup

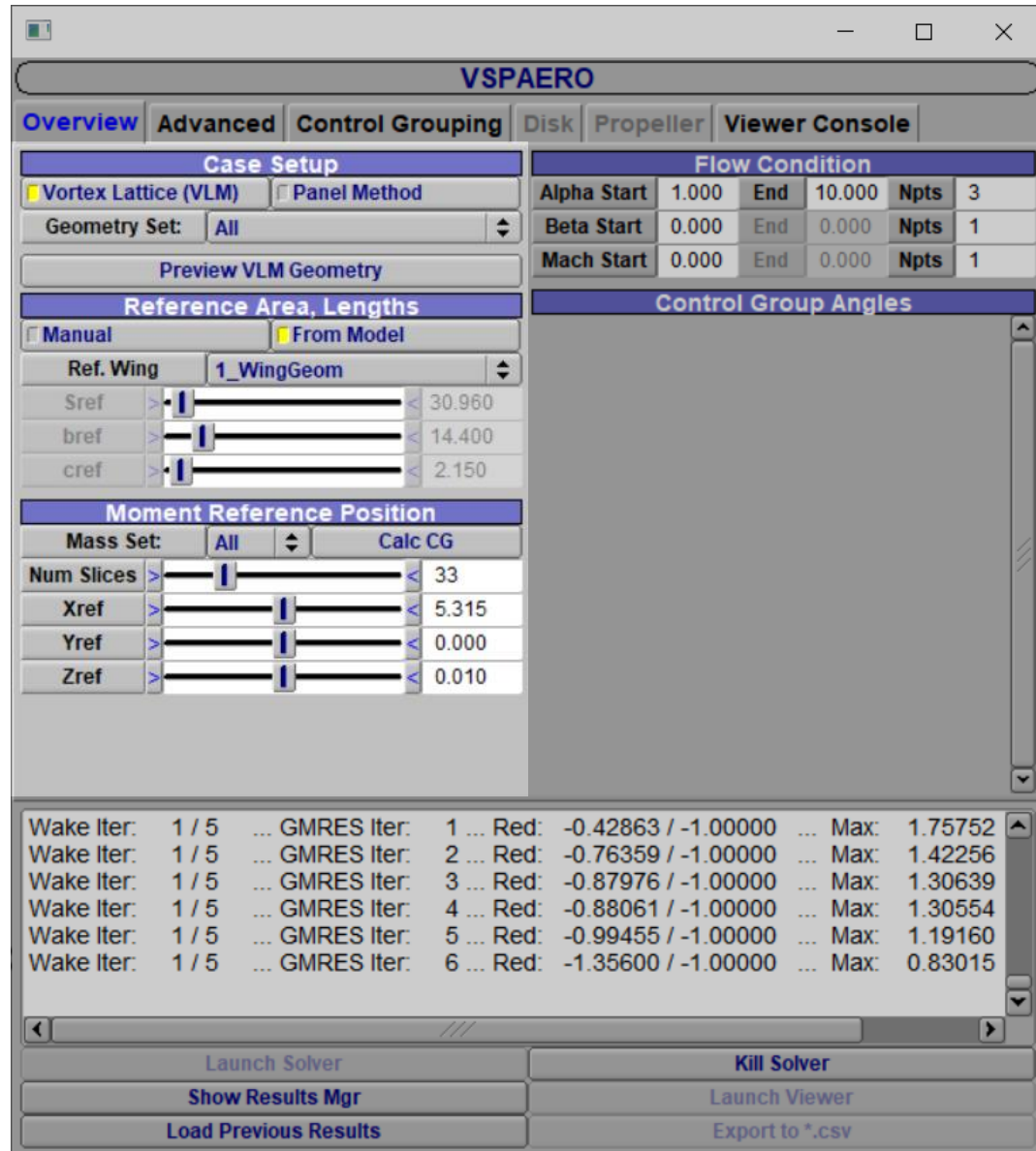
- Analysis methods: VLM or Panel
 - VLM recommended for most cases
- Selectable Geometry Set
- Option to preview VLM input DegenGeom
 - Use Surface dropdown in Geom Browser to change back

Ref Aero, Lengths

- Manual or From Model (wing geom)

Moment Reference Position

- Manual or Calc CG (Mass Properties tool)



VSPAERO

Overview | Advanced | Control Grouping | Disk | Propeller | Viewer Console

Case Setup

☒ Vortex Lattice (VLM) ☐ Panel Method

Geometry Set: All

Preview VLM Geometry

Reference Area, Lengths

☐ Manual ☒ From Model

Ref. Wing: 1_WingGeom

Sref: 30.960
bref: 14.400
cref: 2.150

Moment Reference Position

Mass Set: All ☒ Calc CG

Num Slices: 33
Xref: 5.315
Yref: 0.000
Zref: 0.010

Flow Condition

Parameter	Start	End	Npts
Alpha	1.000	10.000	3
Beta	0.000	0.000	1
Mach	0.000	0.000	1

Control Group Angles

Iteration	Wake Iter	GMRES Iter	Red	Max
1	1 / 5	1	-0.42863 / -1.00000	1.75752
2	1 / 5	2	-0.76359 / -1.00000	1.42256
3	1 / 5	3	-0.87976 / -1.00000	1.30639
4	1 / 5	4	-0.88061 / -1.00000	1.30554
5	1 / 5	5	-0.99455 / -1.00000	1.19160
6	1 / 5	6	-1.35600 / -1.00000	0.83015

Launch Solver | Kill Solver

Show Results Mgr | Launch Viewer

Load Previous Results | Export to *.csv

GUI – Overview Tab (2/2)

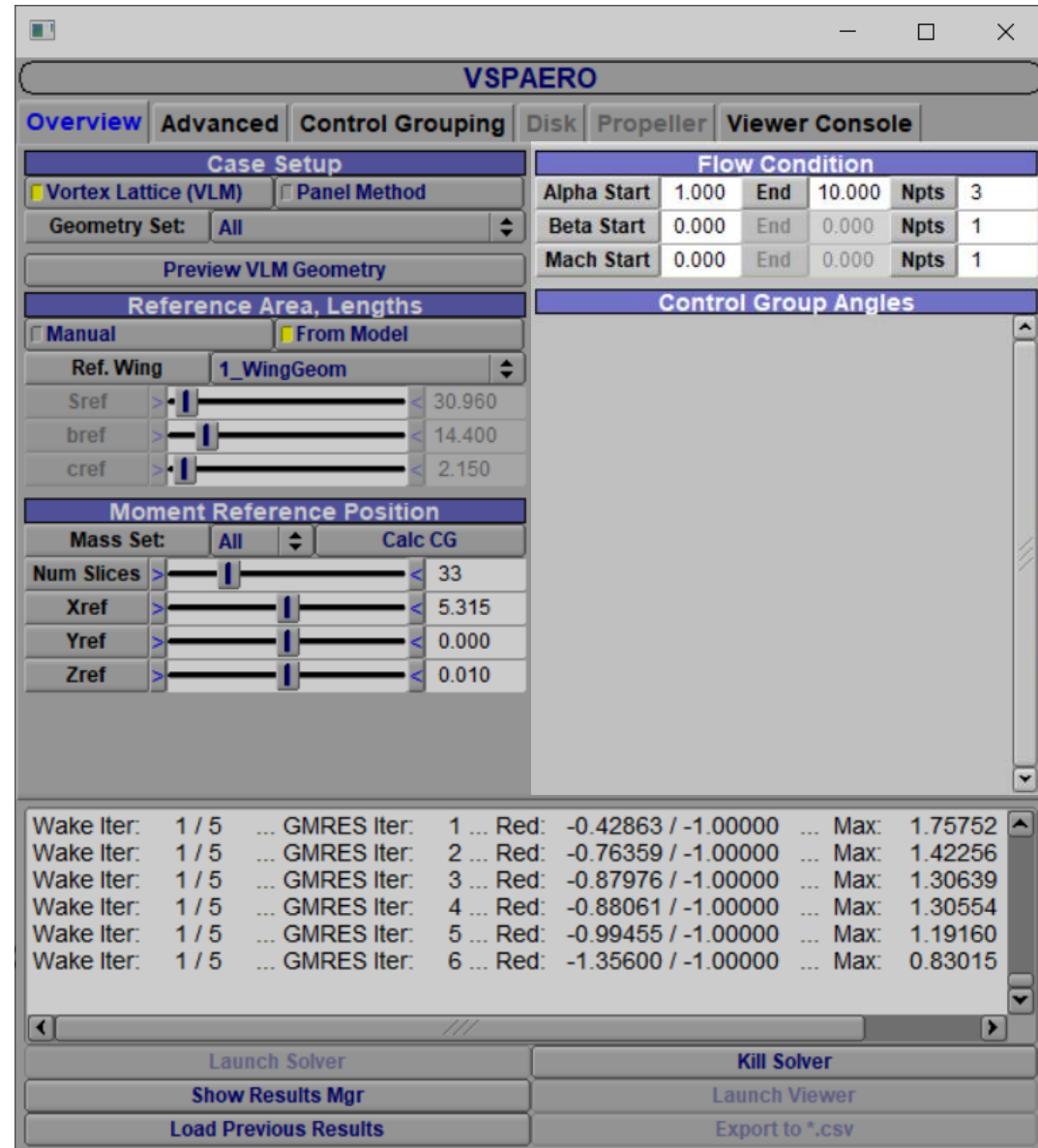
Flow Condition

- Alpha, Beta in degrees
- Avoid transonic Mach values
- Ability to run linearly spaced array of points
 - Multiple flow conditions not supported with some advanced options (i.e. unsteady analysis)

Control Group Angles*

- Enable\disable control groups
- Set deflection angle (deg)

*See “VSPAERO GUI Advanced” presentation



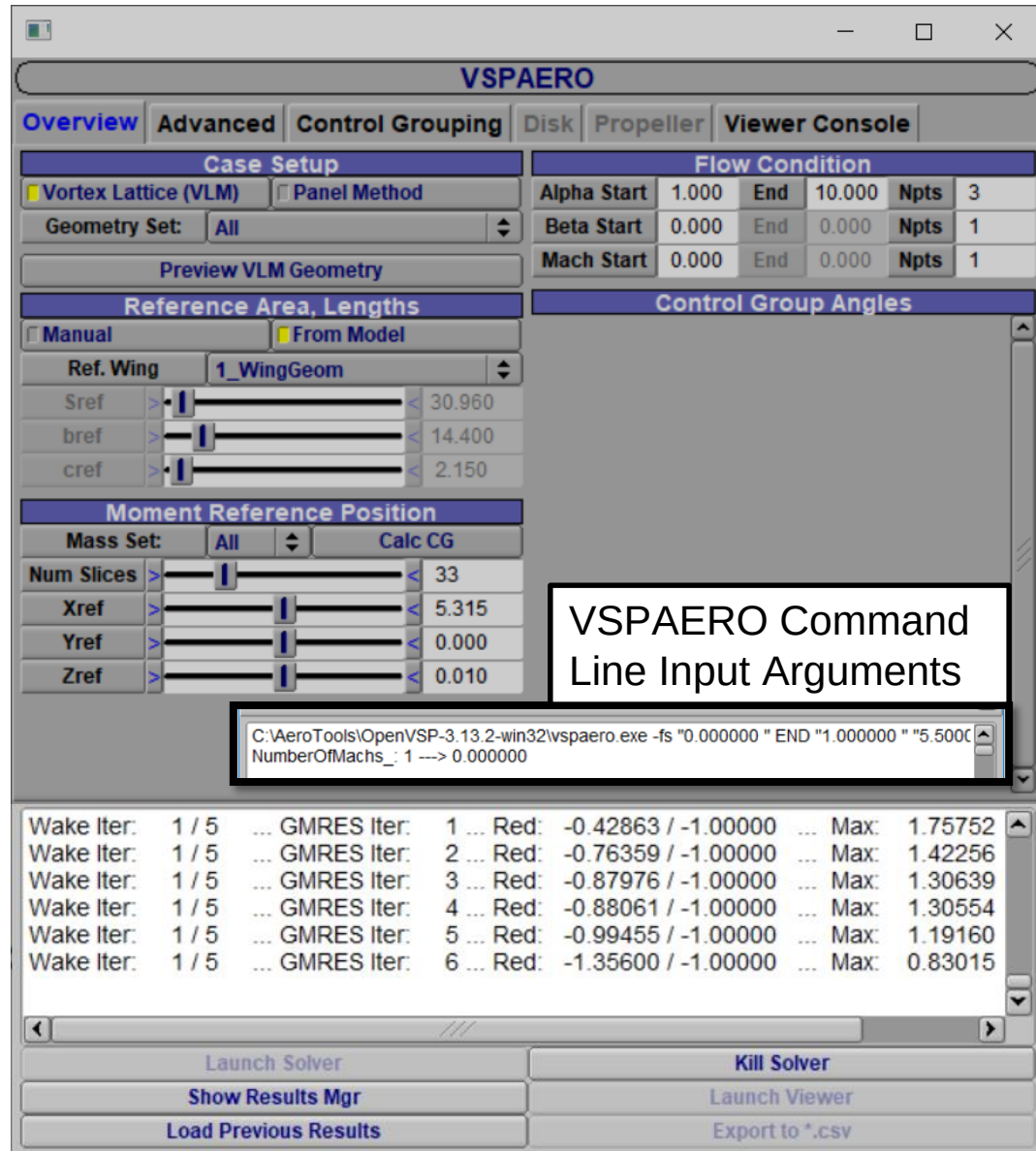
GUI – Solver Console & Execution

Solver Console

- VSPAERO status messages
- Solver console always visible
- VSPAERO command arguments shown first

Execution

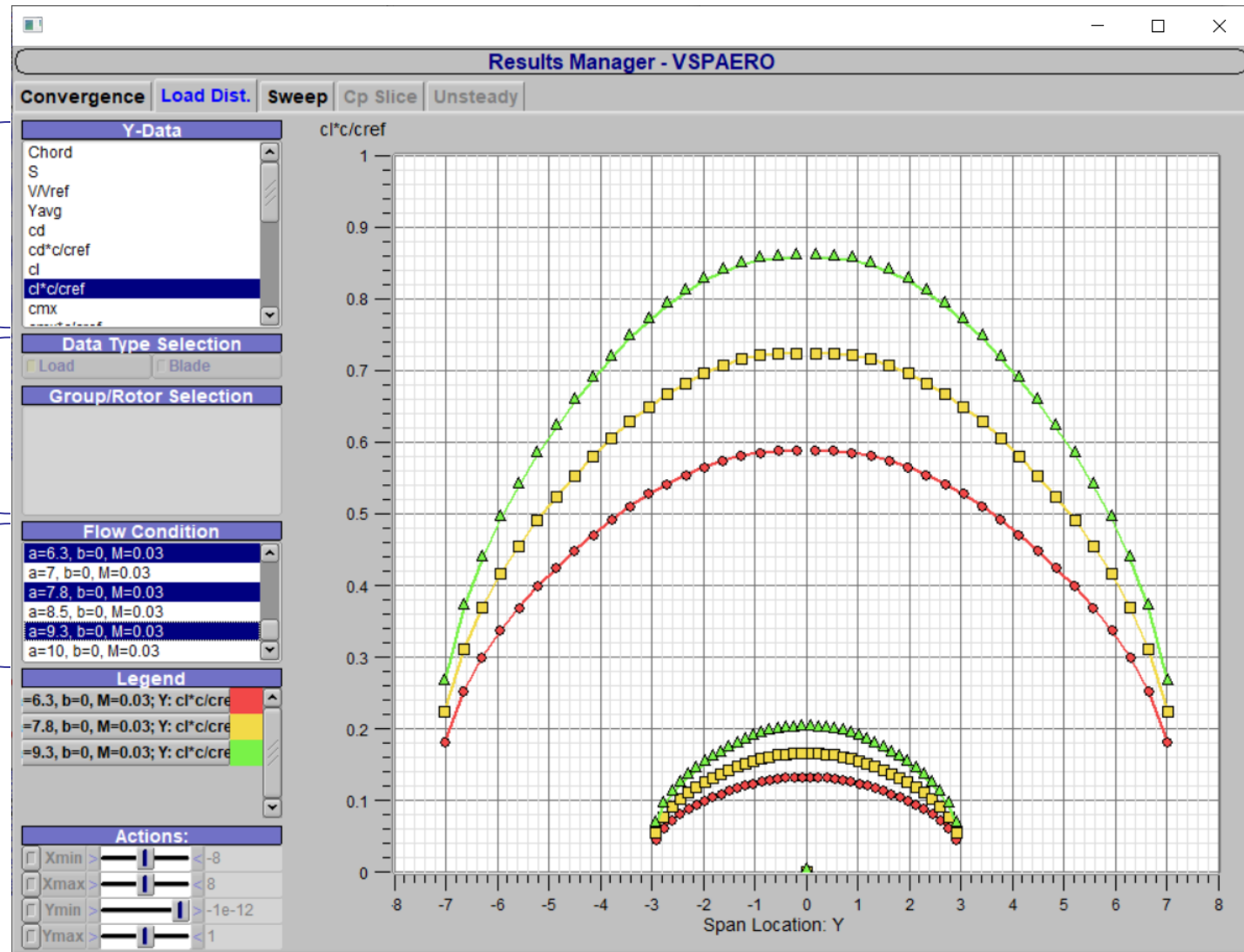
- Launch Solver to execute VSPAERO
 - Kill Solver to terminate prematurely
- Results Manager Plot Screen & VSPAERO Viewer available after execution
- *New Feature:* Load results from previous VSPAERO run
 - Looks for files with base DegenGeom/Cart3D name



VSPAERO Command
Line Input Arguments

GUI Overview – Results Manager: Load Dist.

- Plots *.lod output file



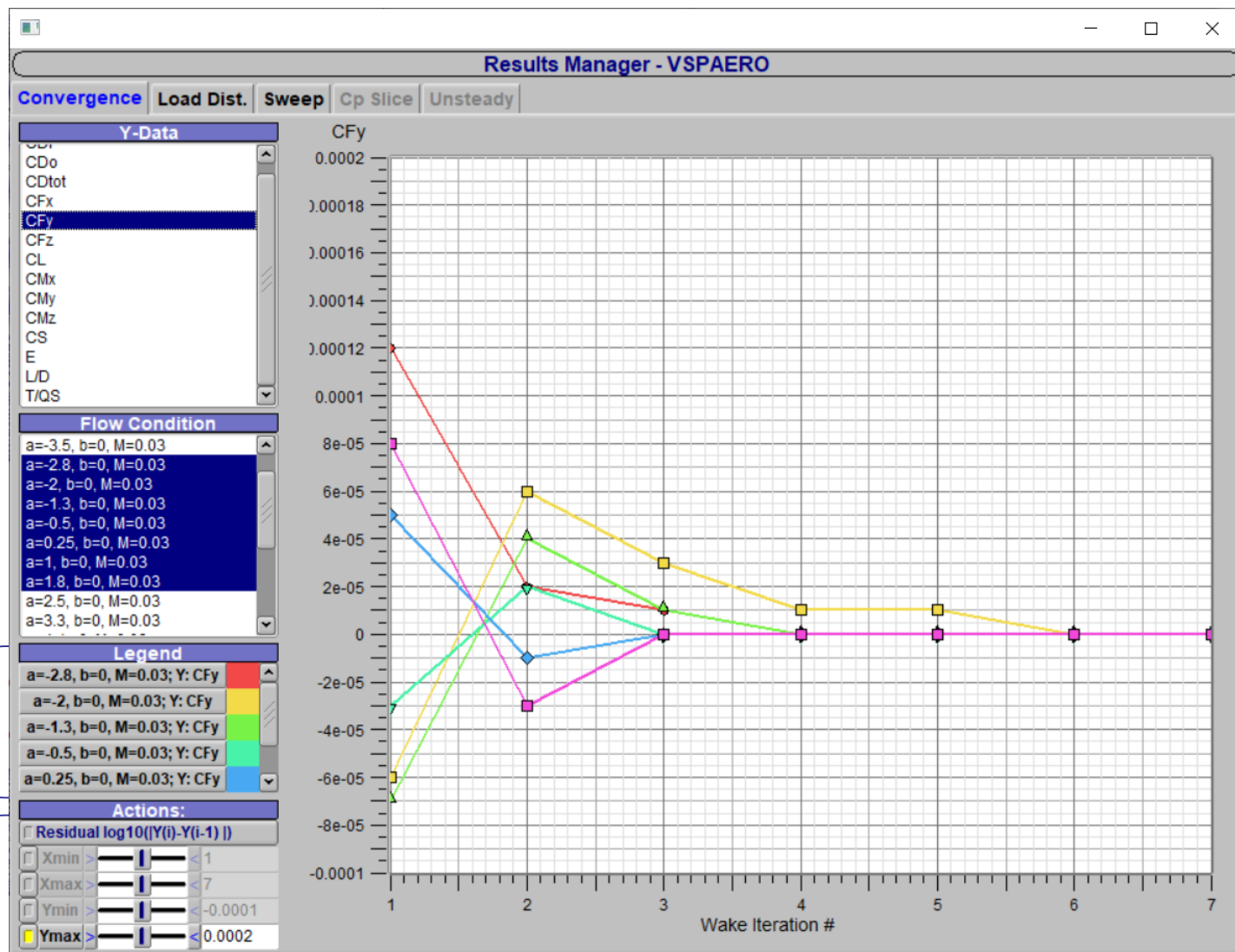
- Highlights spanwise distribution of results

GUI Overview – Results Manager: Convergence

- Plots *.history output file
- Highlights changes across wake iterations
- Optional $\log_{10}()$ of residual
 - If difference between iterations

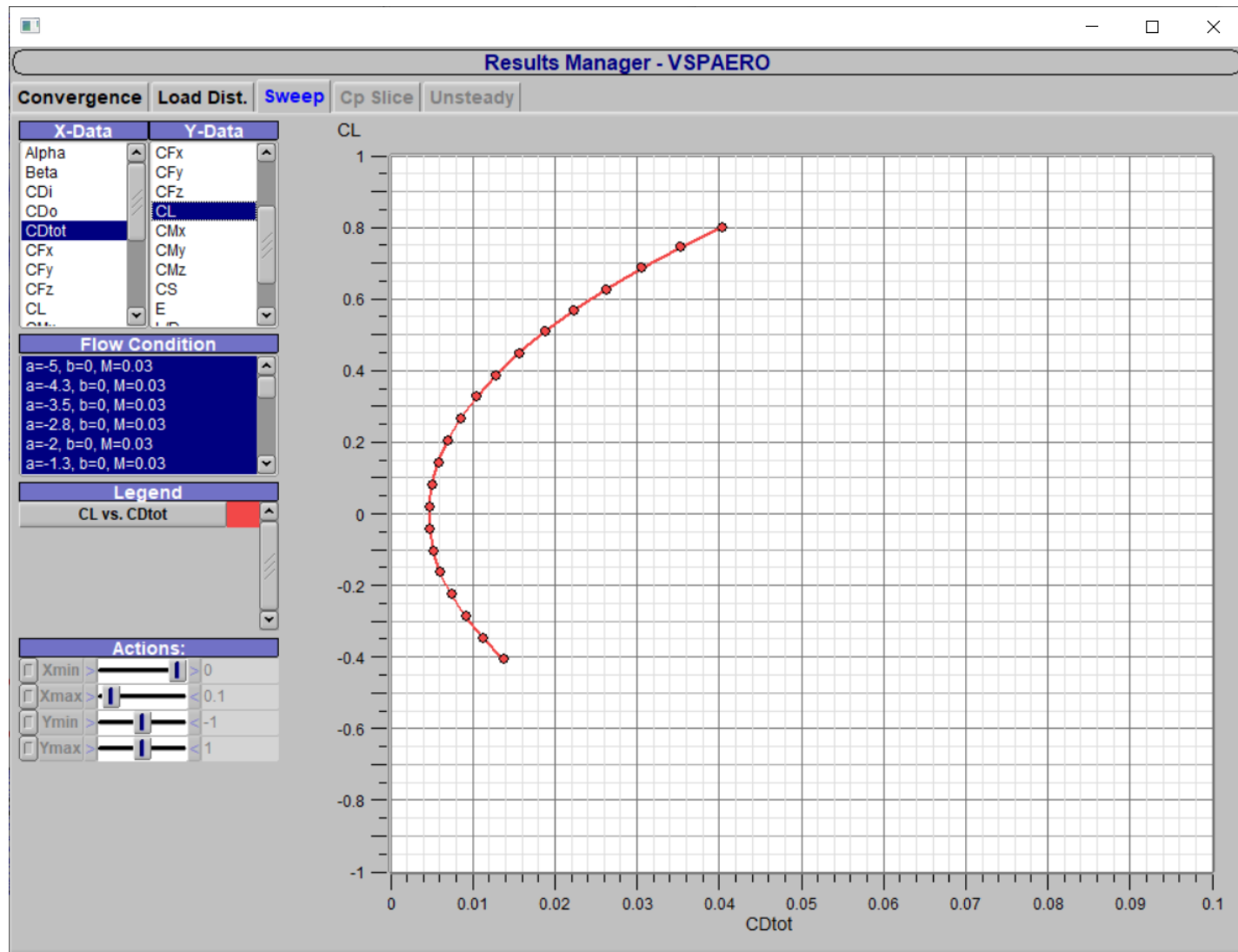
Legend

Manual Axis Limits

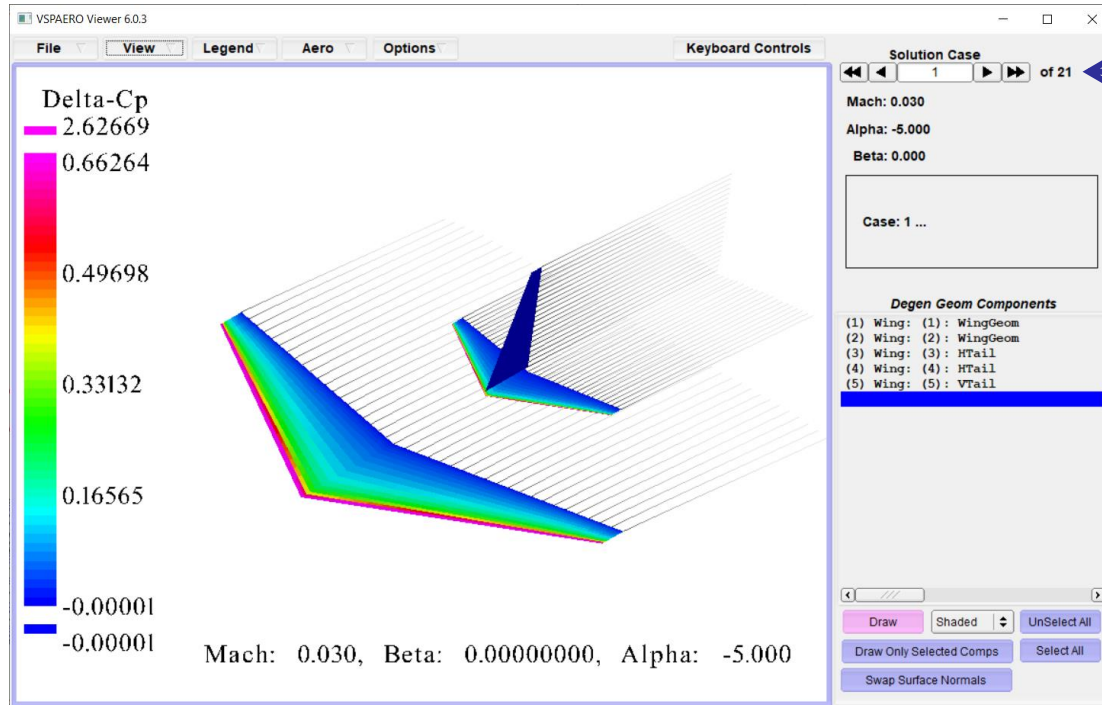


GUI Overview – Results Manager: Sweep

- Plots *.history output file
- Result from last wake iteration
- Highlights changes across flow conditions – i.e. drag polar
- Multiple data selection is supported for X or Y (not both)



VSP Viewer



Step Through Flow Condition or Time (Unsteady)

Component List and Drawing Settings

☒ Use Per Tri Function Shading
☐ Use Smooth Function Shading
☐ Use Line Contours

Function Contour Type

☒ Off
☐ On
☐ On _Deflected

Draw Beam3D Model

☐ Draw X-Plane
☐ Draw Y-Plane
☐ Draw Z-Plane

Draw Axis Planes

☒ Off
☐ Colored by Wing ID
☐ Colored by Span Location

Draw Axis Planes

Reset View	Ctrl+R
Top View	Ctrl+F1
Bottom View	Ctrl+F2
Left View	Ctrl+F3
Right View	Ctrl+F4
Front View	Ctrl+F5
Rear View	Ctrl+F6
Iso View Up	Ctrl+F7
Iso View Down	Ctrl+F8
☐ Use Blue Background	

View

☐ Run Label
☐ Contours Legend
☐ Draw CG
☐ Draw CG Label Info
☐ Draw Axes

Legend

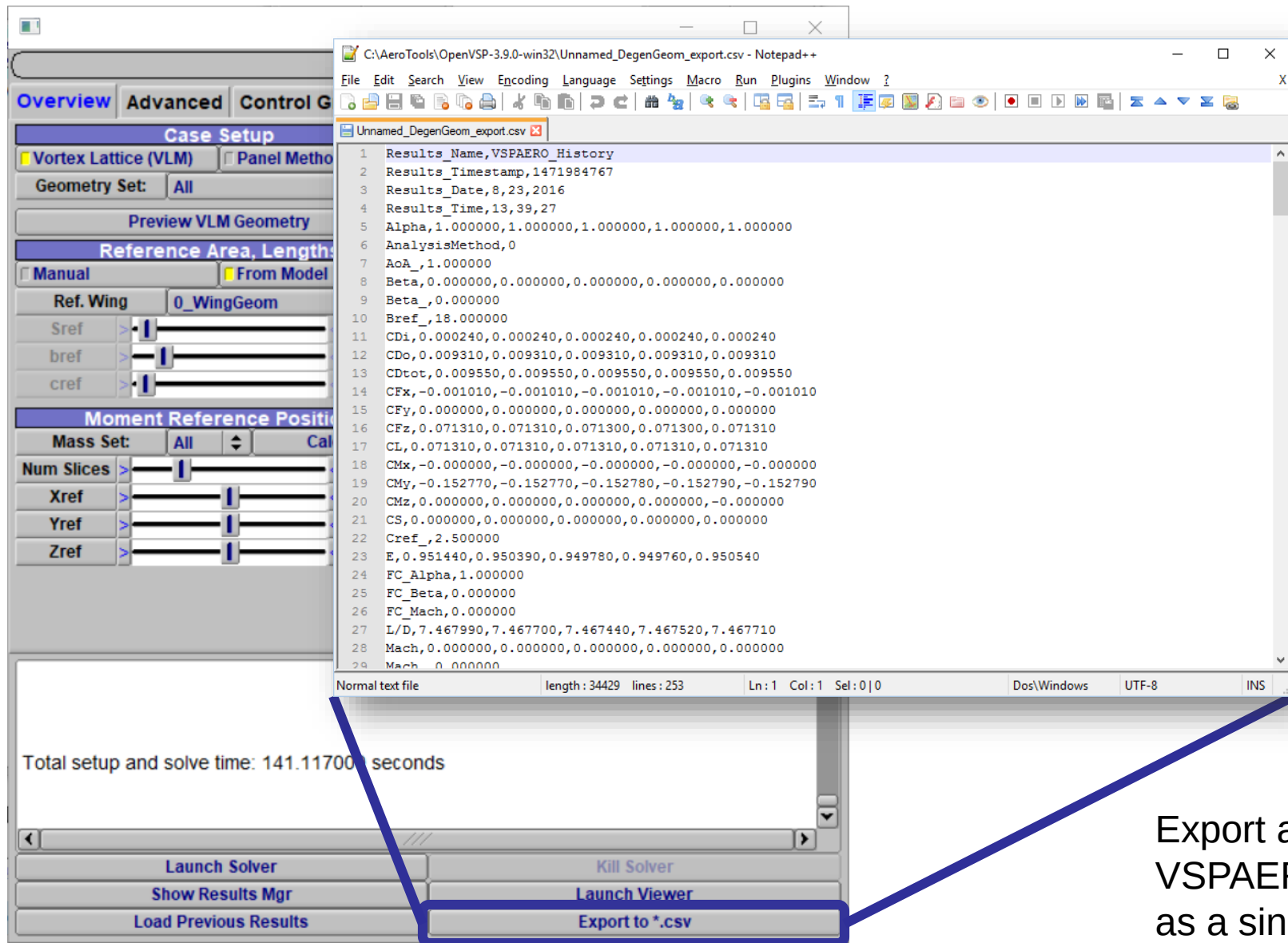
☒ Shaded
☐ Pressures
☐ Steady Pressures
☐ Unsteady Pressures
☐ Vorticity
☐ Triangulation
☐ Computational Mesh
☐ Trailing Wakes
☐ Rotors
☐ Control Surfaces

Aero

Set Contour Levels
Function Contour Type
☐ Use Flat Shading
☐ Draw Wakes To Infinity
☐ Draw Control Surfaces Deflected
☐ Draw Reflected Geometry
☒ Draw Wakes As Lines
☐ Draw Wakes As Points
Draw Beam3D Model
Draw Axis Planes
Draw Colored Wakes

Options

GUI Overview – Export



The screenshot shows the VSPAERO GUI with the 'Export to *.csv' button highlighted. An arrow points from this button to a Notepad++ window displaying the exported CSV data. The CSV data includes the following fields:

```

1 Results_Name,VSPAERO_History
2 Results_Stamp,1471984767
3 Results_Date,8,23,2016
4 Results_Time,13,39,27
5 Alpha,1.000000,1.000000,1.000000,1.000000,1.000000
6 AnalysisMethod,0
7 AoA,1.000000
8 Beta,0.000000,0.000000,0.000000,0.000000,0.000000
9 Beta_,0.000000
10 Bref,18.000000
11 CDi,0.000240,0.000240,0.000240,0.000240,0.000240
12 CDo,0.009310,0.009310,0.009310,0.009310,0.009310
13 CDtot,0.009550,0.009550,0.009550,0.009550,0.009550
14 CFx,-0.001010,-0.001010,-0.001010,-0.001010,-0.001010
15 CFy,0.000000,0.000000,0.000000,0.000000,0.000000
16 CFz,0.071310,0.071310,0.071300,0.071300,0.071310
17 CL,0.071310,0.071310,0.071310,0.071310,0.071310
18 CMx,-0.000000,-0.000000,-0.000000,-0.000000,-0.000000
19 CMy,-0.152770,-0.152770,-0.152780,-0.152790,-0.152790
20 CMz,0.000000,0.000000,0.000000,0.000000,-0.000000
21 CS,0.000000,0.000000,0.000000,0.000000,0.000000
22 Cref,2.500000
23 E,0.951440,0.950390,0.949780,0.949760,0.950540
24 FC_Alpha,1.000000
25 FC_Beta,0.000000
26 FC_Mach,0.000000
27 L/D,7.467990,7.467700,7.467440,7.467520,7.467710
28 Mach,0.000000,0.000000,0.000000,0.000000,0.000000
29 Mach_ 0.000000
  
```

The GUI also shows the 'Total setup and solve time: 141.11700 seconds' and the 'Export to *.csv' button highlighted with a blue box.

Export all
VSPAERO results
as a single CSV

GUI Overview – Basic Workflow



1

1. Analysis → VSPAERO

2. Select options

3. [Optional] Preview VLM Geom

4. Launch/Kill Solver

5. View & Export Results

- Show Results Mgr (launched automatically after execution)
- Launch Viewer
- Export to *.csv

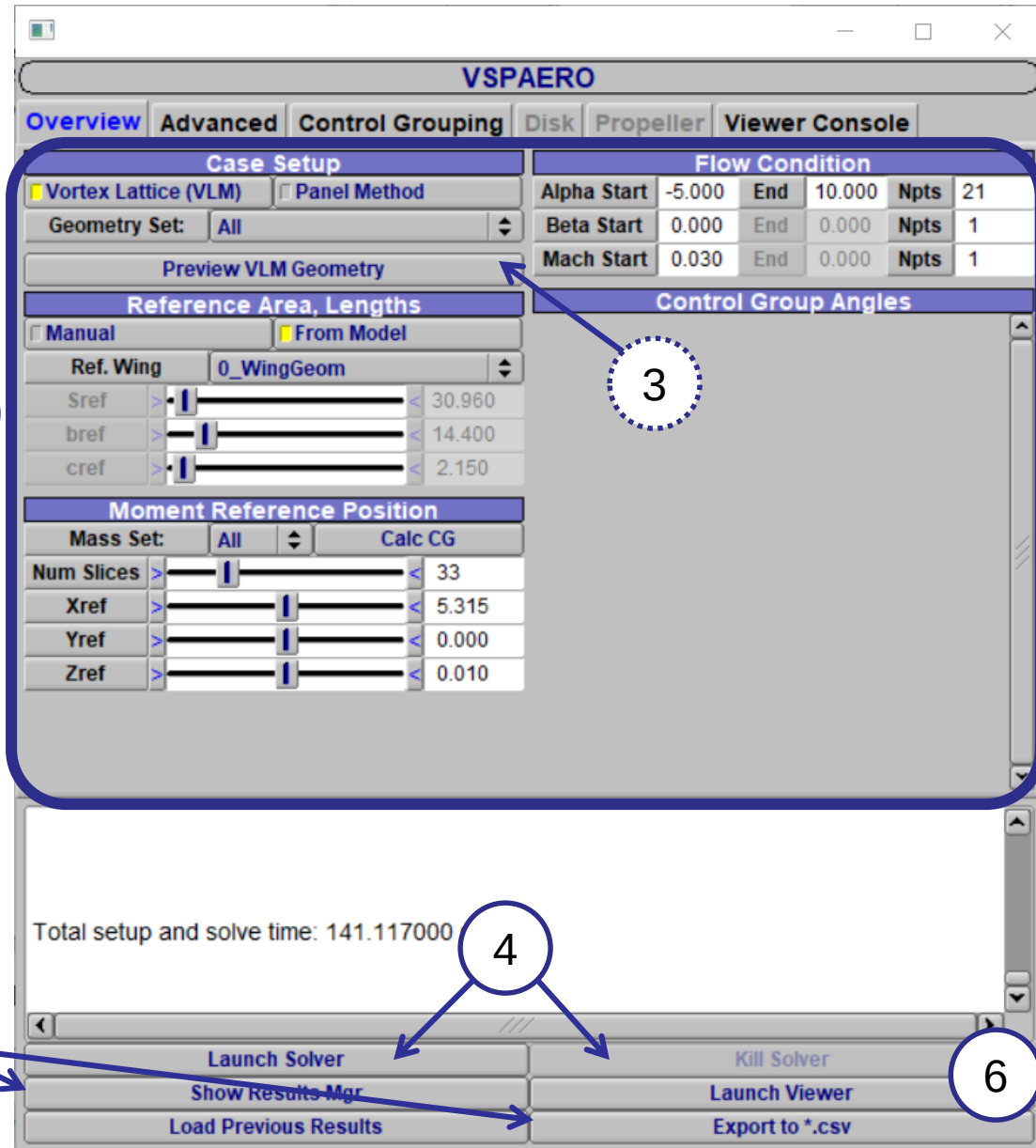
2

3

4

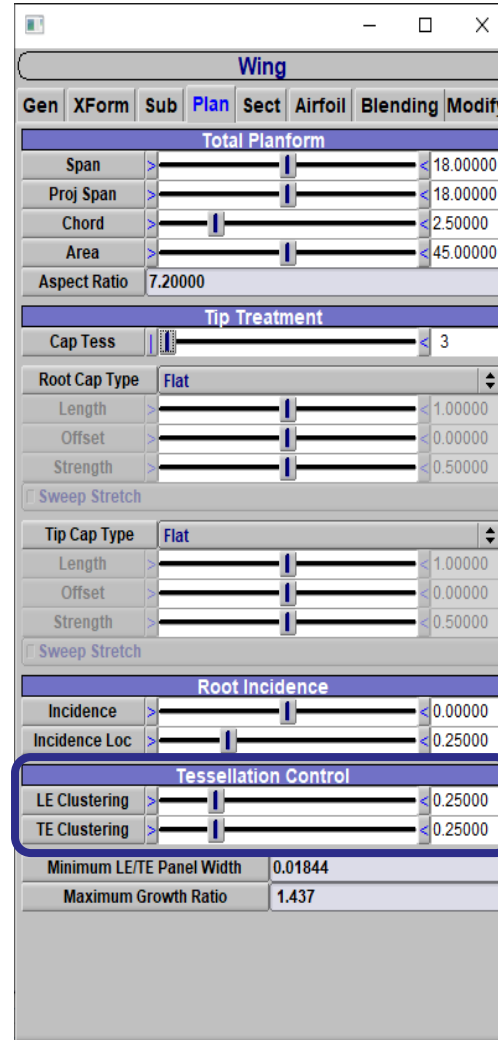
5

6



Tessellation

- Wing
 - Control for LE/TE and root/tip clustering
 - Chordwise (W): “Plan” tab
 - Spanwise (U): “Sect” tab
 - Number and location of wakes
- Bodies
 - Skinning controls clustering
- Increased tessellation:
 - Increased computing time
 - Usually higher fidelity – not always
- Adjusting geometry or tessellation slightly can fix mesh and solution issues



Wing

Gen XForm Sub **Plan** Sect Airfoil Blending Modify

Total Planform

Span	18.0000
Proj Span	18.0000
Chord	2.5000
Area	45.0000
Aspect Ratio	7.2000

Tip Treatment

Cap Tess 3

Root Cap Type Flat

Length	1.0000
Offset	0.0000
Strength	0.5000

☐ Sweep Stretch

Tip Cap Type Flat

Length	1.0000
Offset	0.0000
Strength	0.5000

☐ Sweep Stretch

Root Incidence

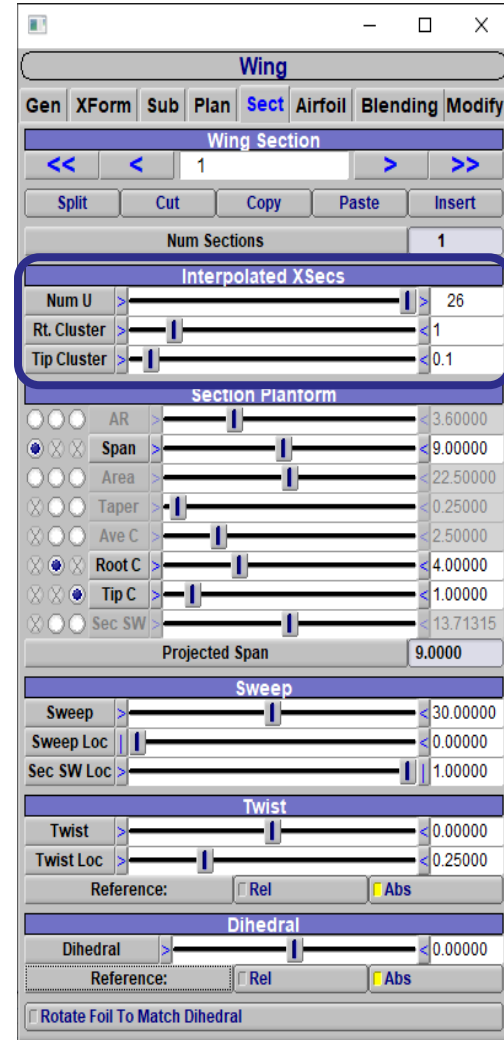
Incidence	0.0000
Incidence Loc	0.2500

Tessellation Control

LE Clustering	0.2500
TE Clustering	0.2500

Minimum LE/TE Panel Width 0.01844

Maximum Growth Ratio 1.437



Wing

Gen XForm Sub Plan **Sect** Airfoil Blending Modify

Wing Section

1

Split Cut Copy Paste Insert

Num Sections 1

Interpolated Xsecs

Num U	26
Rt. Cluster	1
Tip Cluster	0.1

Section Planform

AR	3.6000
Span	9.0000
Area	22.5000
Taper	0.2500
Ave C	2.5000
Root C	4.0000
Tip C	1.0000
Sec SW	13.71315

Projected Span 9.0000

Sweep

Sweep	30.0000
Sweep Loc	0.0000
Sec SW Loc	1.0000

Twist

Twist	0.0000
Twist Loc	0.2500

Reference: ☐ Rel ☒ Abs

Dihedral

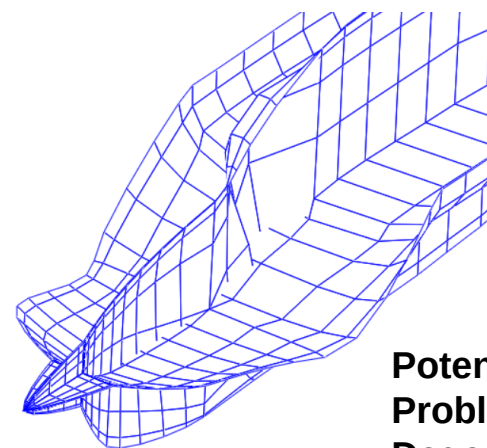
Dihedral	0.0000
----------	--------

Reference: ☐ Rel ☒ Abs

☐ Rotate Foil To Match Dihedral

Modeling Tips for VSPAERO

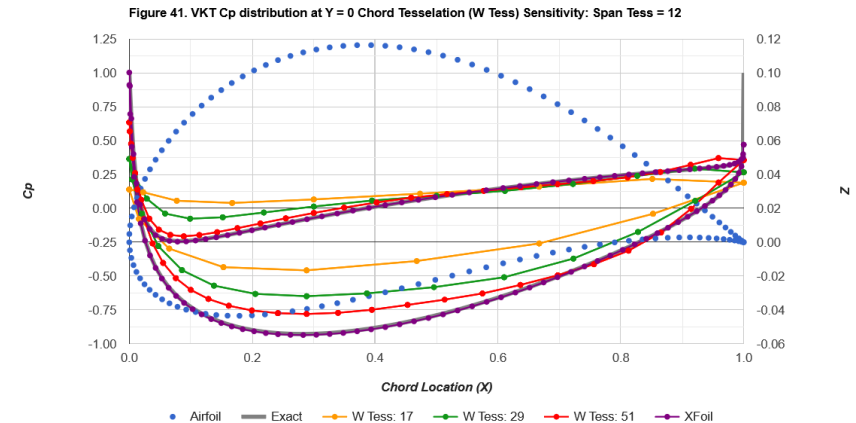
- Start simple
 - Few wake iterations
 - Simple geometries
 - VLM over Panel Method
- Check VLM geometry preview →
- Ensure components are closed
 - Model 1st and last fuselage XSec as point type
- U tessellation sets number of wakes for lifting surfaces
 - Increase tessellation -> increase execution time
 - Increasing tessellation usually improves fidelity – use caution
- Be careful including body components in VLM analysis
 - Do not contribute to lift or induced drag
 - May introduce errors when modeled incorrectly



**Potentially
Problematic
DegenGeom**

Modeling Tips for VSPAERO

- Do not use rounded rectangle airfoils – incompatible with VSPAERO
- Avoid high U (spanwise) with low W (chordwise) tessellation or vice versa
- Slight adjustments to geometry or tessellation can alleviate mesh and solution issues
- Several wiki pages covering VSPAERO
- Post questions and issues to the Google group



<http://openvsp.org/wiki/doku.php?id=vspaerosummary>



OpenVSP

Trace: • [workshop2019](#) • [cfdmesh](#) • [feamesh](#) • [modelingguides](#) • [start](#) • [analysisguides](#) • [vspaer](#)

Guides to VSPAERO

[Modeling for VSPAERO](#)

[VSPAERO Tutorial](#)

[VSPAERO Common Problems](#)

[CpSlicer](#)

Demo